

The Collapse Operator

$$(\hat{\Omega})$$

The Universal Selection Mechanism for Emergent Constants

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EI Tag

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Tagline: The Legend of the Atlas

Symbol: $\hat{\Omega}$ — Recursive Collapse / Viability Operator ($\zeta-\omega-\xi$)

Collapse Consequence:

If $\hat{\Omega}$ cannot be stated consistently across domains (or yields contradictions), the Atlas reduces to disconnected curve-fits and the 7dU survivability program fails.

AGI Verifier Hint:

Start here if you are auditing whether “constants as survivors” is a coherent program.

Verify that $\hat{\Omega}$ can be written explicitly in multiple domains (geometry, causality, coupling, confinement), that its specialization rules prevent unit-circularity, and that alternative operator families can be discriminated by falsifiers.

Abstract

The Collapse Operator ($\hat{\Omega}$) is the foundational selection mechanism of the Atlas of Constants (AoC) program. It governs the emergence of physical constants (π , c , α , m_p , ...) as stable eigenvalues or fixed points within recursive geometric systems subject to constraint (ζ), extension (ω), and chance (ξ).

This entry establishes the canonical form of $\hat{\Omega}$ in three equivalent representations—spectral, recursive, and variational—and defines the epistemic contract for all downstream constant derivations. By normalizing geometry to dimensionless ratios (separating structure from units) and testing persistence under entropy injection, the operator yields universal survivors: constants that cannot be further collapsed without breaking coherence.

$\hat{\Omega}$ is presented as [ANSATZ]: the axiomatic primitive upon which the Atlas is built. All derived constants inherit this status conditionally ([DERIVED given $\hat{\Omega}$]). The framework is falsified if: (1) different constants require contradictory operator structures, (2) the operator cannot be written explicitly in a domain, or (3) it yields internal contradictions among constants.

1. Canonical Forms of $\hat{\Omega}$

To ensure applicability across diverse physical domains (geometry, causality, coupling, confinement), we define the Collapse Operator $\hat{\Omega}$ in three representations. These are claimed-to-be-equivalent in the sense made explicit in Section 1.4 (Equivalence Bridge); downstream AoC entries must declare which representation they employ and which bridge conditions they rely on.

1.1 Recursive / Viability Face (CCT-aligned)

This representation models $\hat{\Omega}$ as an iterative test of structural persistence. It is the primary form for reasoning about identity survival, threshold behavior, and classification under repeated ζ - ω - ξ stress.

Definitions:

- M_n : system state / configuration at recursion depth n . (Domain examples: iteration count in numerical solvers; renormalization steps; collapse epochs; etc.)
- $G(\cdot)$: geometry extraction map. It maps M_n to the domain's geometric descriptor(s): a scalar invariant, a spectrum, a tensor field, and/or boundary/admissibility data.
- $\mathcal{N}_\xi$: entropy injection field (structured stochastic load) governed by ξ .
- $R(\cdot, \cdot)$: viability map returning {Persist, Collapse, Diverge}.

$$\hat{\Omega}(M_n) := R(G(M_n), \mathcal{N}_\xi)$$

Operational definition (domain-agnostic):

- Persist: bounded deviation with a stable attractor class / invariant measure under iteration.
- Collapse: convergence to a degenerate / trivial state (loss of degrees of freedom, null geometry, etc.).

- Diverge: unbounded growth or bounded-but-nonconvergent behavior that fails to project onto a stable invariant measure.

Note (edge case discipline): oscillatory/chaotic bounded behavior is classified as Diverge unless the deriving entry demonstrates a stable invariant measure that satisfies the Persist criterion.

Status: [ANSATZ]

1.2 Spectral / Operator Face (dimensionless eigen-structure)

This representation expresses $\hat{\Omega}$ as an operator on a declared domain with declared boundary conditions and admissible function space. It is the primary form for deriving dimensionless constants as stable spectral invariants.

$$\backslash\boxed{\quad \hat{\Omega}_{\zeta,\omega,\xi} := s_{\text{geom}} \left[-\zeta \nabla^2 + \omega \mathcal{R}[\cdot] + \xi \mathcal{P}(x) \right]}$$

Definitions:

- $-\zeta \nabla^2$: constraint term (locality / confinement; suppresses divergence).
- $\omega \mathcal{R}[\cdot]$: extension / persistence feedback term; $\mathcal{R}$ is domain-declared (curvature functional, response functional, etc.).
- $\xi \mathcal{P}(x)$: chance injection class; $\mathcal{P}$ is domain-declared (white/colored/bounded/structured; see §6.2).
- s_{geom} : normalization ensuring outputs are dimensionless invariants.

Choice must be declared and audited; alternatives are enumerated in §6.2.

Canonical example (non-exclusive): choose s_{geom} as an inverse reference scale built from the domain's baseline operator magnitudes, e.g.

$$s_{\text{geom}} := \frac{1}{\lambda_{\min}(-\nabla^2) + \omega \mathcal{R}_0}$$

where

$\mathcal{R}_0$ is a declared baseline reference value for the domain (or its appropriate invariant surrogate). This example exists to anchor auditability; it does not restrict the allowed normalization family.

Use constraint: downstream entries must specify whether they are solving an eigenproblem

$$\hat{\Omega} \psi_k = \lambda_k \psi_k$$

or an evolution equation driven by $\hat{\Omega}$ (if used, the evolution law must be explicitly stated).

Status: [ANSATZ]

1.3 Variational Face (domain-dependent extremum principle)

Where the domain admits a well-defined action/functional, $\hat{\Omega}$ may be represented as a variational constraint. This representation is not assumed universal; it is domain-dependent.

\boxed.

$$\delta \mathcal{V}[M] = 0$$

A candidate structure (illustrative, not canonical) is:

$$\mathcal{V}[M] := \int_{\mathcal{D}} \left(\zeta \|\nabla M\|^2 + \omega \mathcal{F}[M] + \Phi[M; \xi, \mathcal{N}_\xi] \right) d\mu$$

where Φ encodes the declared entropy-tolerance or noise-response penalty/constraint class.

Status: [CONJECTURE]

(until a domain-explicit functional is provided and shown to reproduce the spectral/recursive behavior)

1.4 Equivalence Bridge (auditable bridge claims)

The Atlas remains coherent only if these representations are related by explicit, testable bridge claims. We state the bridge conditions as claims with verification obligations, not as assumed facts.

Bridge Claim 1 — Noise correspondence:

$\mathcal{N}_\xi$ in the recursive form corresponds to the declared stochastic class $\mathcal{P}(x)$ in the operator form, under a domain-declared mapping.

Mapping (schematic): $\mathcal{N}_\xi \leftrightarrow \xi \mathcal{P}(x)$.

Bridge Claim 2 — Geometry/feedback correspondence:

$G(M_n)$ supplies the geometric descriptor(s) that determine $\mathcal{R}[\cdot]$ and the boundary/admissibility structure.

Mapping (schematic): $G(M) \rightsquigarrow \mathcal{R}[\cdot]$ plus boundary conditions.

Bridge Claim 3 — Selection correspondence:

A state classified as Persist by the viability map corresponds to stability in the operator representation (stable eigenvalue(s) under the declared perturbation class and normalization).

Operational test: stability under iteration in 1.1 must match stability of the relevant spectral invariant(s) in 1.2 within declared tolerances.

Disclosure clause: equivalence may be partial or regime-limited; any gap must be explicitly tagged in the deriving entry.

Forward references (first uses): see Appx_7_AoC_0_On_Pi (bridge use on closure/eigenstructure) and AoC_1_On_c (bridge use on causal/null-slope specialization) as the first two domain instantiations.

Status: [CONJECTURE]

(bridge claims) — verified or falsified case-by-case in downstream entries.

2. Operator Semantics (ζ , ω , ξ)

This section defines the canonical semantics of the parameters used in the spectral/ operator face (and referenced by the recursive and variational faces via the bridge claims). These meanings are not re-derived in downstream AoC entries; instead, entries specialize them by declaring domain, boundary conditions, admissible function spaces, and noise class.

2.1 Canonical table (used everywhere)

Term	Meaning (7dU / AoC)	Mathematical action (operator face)	Failure mode if removed / degenerate
ζ	Constraint (locality / boundedness / anti-blowup)	Suppresses divergence; enforces locality via a confining operator (canonical: $-\zeta \nabla^2$; allowed families enumerated in §6.2).	Runaway modes / loss of locality; “instantaneous action-at-a-distance” pathologies in causal domains; non-survivable recursion (no bounded attractors).
ω	Extension (persistence / continuity / long-mode support)	Couples to a persistence/ feedback functional (canonical: $+\omega \mathcal{R}[\cdot]$) that sustains nontrivial structure across domain scale.	Freeze-out to triviality or fragmentation into disconnected micro-structures; loss of large-scale / long-range coherence; no durable invariants.
ξ	Chance (entropy sampling / novelty / selection pressure)	Injects stochastic load through a declared perturbation class (canonical: $+\xi \mathcal{P}(x)$, domain-declared).	Sterile determinism; degeneracy among candidate survivors is unresolved; no selection mechanism under perturbation.
s_{geom}	Scale removal (dimensionless invariants; audit discipline)	Normalization that maps the operator outputs to dimensionless quantities; choice must be declared and audited (canonical example in §1.2; alternatives in §6.2).	Unit leakage; circular “calibration” risk; invariants become coordinate/metrology artifacts rather than structural survivors.

3. Domain-Agnostic Outputs

When the Collapse Operator acts on a declared domain state M (via recursion, spectral analysis, or a variational formulation), outcomes fall into a small set of audit-stable behavioral regimes. These classes are intended to be domain-agnostic; the exact norm/metric, tolerances, and thresholds must be declared by the deriving entry.

3.1 Output classes (regimes)

Let $d(\cdot, \cdot)$ be the declared distance/metric on the admissible state space, and let M_n denote the iterated state under the chosen operator face (or solver evolution).

Regime	Operational signature (must be declared)	Structural meaning
Divergence	$\ M_n\ \rightarrow \infty$, or $d(M_{n+1}, M_n)$ exceeds a declared threshold (runaway, instability, uncontained growth).	Non-survivable recursion / non-physical blow-up for the declared domain.
Collapse	$M_n \rightarrow M_\emptyset$ where $M_\emptyset$ is the declared degenerate class (zero field, trivial geometry, singular pinching, loss of degrees of freedom), or invariants collapse to trivial/undefined values.	Degeneration: the system cannot sustain nontrivial structure under the declared ζ, ω, ξ and boundary conditions.
Coherence (<i>Survivor class</i>)	Existence of a stable invariant object under iteration: a fixed point M^* , a stable eigenmode, or a stable invariant measure. Concretely: boundedness + reproducible convergence of a declared invariant (e.g., eigenvalue) within tolerance under variation of initial conditions, random seeds, and/or discretization/refinement (as applicable to the domain).	Survivable structure: a dimensionless invariant persists under the declared stress (constraint/extension/chance).

Oscillation/Chaos Clause: bounded-but-nonconvergent behavior (limit cycles, quasi-periodic motion, bounded chaos) is classified as Divergence unless the deriving entry demonstrates convergence to a stable invariant measure or stable projected invariant (in which case it is Coherence).

Status: [DERIVED given $\hat{\Omega}$]

3.2 Survivor definition (CCT-1 interface)

A “Survivor” is the object of interest in the Atlas: a persistent invariant produced by $\hat{\Omega}$ in a declared domain.

Depending on the operator face used (§1), “Survivor” means:

- Spectral face: a stable eigenfunction ψ_* with a real, dimensionless eigenvalue λ_* (the candidate constant or invariant ratio), stable under the declared perturbation class and discretization refinement. (If a domain admits negative λ_* , this must be stated; otherwise positivity may be assumed by the deriving entry.)
- Recursive face: a fixed point M_* or stable invariant measure such that the declared invariant(s) remain bounded and reproducible under $\mathcal{N}_\xi$ (entropy injection) and iteration depth.
- Variational face (domain-dependent): a stationary configuration/path M_* that extremizes (typically minimizes) the declared viability functional $\mathcal{V}[M]$ under admissible variations.

CCT-1 alignment (language bridge): the Survivor corresponds to a persistent identity $\mathcal{J}$ across recursion depth—an irreducible structure that remains stable under the declared survivability constraints (see Appendix 16: CCT-1).

Guard clause (non-negotiable): Persistence indicates viability, not absolute truth. A Survivor is “what can stably exist under the declared assumptions,” not “what must be unique” and not “what is automatically correct.” Uniqueness claims must be argued explicitly (e.g., by degeneracy-breaking, discriminator tests, or cross-domain consistency).

Status: [DERIVED given ANSATZ (CCT-1 interface)]

4. Atlas Contract

This section is the binding contract for all Atlas of Constants (AoC) entries. Its purpose is to prevent hidden assumptions, post-hoc calibration, and rhetorical bridging. AoC claims must remain auditable, falsifiable, and collapse-resilient.

4.1 Epistemic dependency rule

Unconditional derivation claims are prohibited. Every result depends on (i) the operator and (ii) domain declarations.

Rule: Every AoC entry must use conditional labeling:

Status: [DERIVED given ANSATZ 1–k]

Requirement: The full ANSATZ list must appear in the header (or Section 0), including:

- operator face invoked (spectral / recursive / variational),
- domain declaration (manifold type, boundary conditions, admissible function space),
- noise class / injection model (if used),
- normalization choice s_{geom} ,
- bridge-claim dependencies (if relying on §1.4 equivalence claims),
- any discretization/refinement assumptions (if numerical).

Status: [DERIVED] (epistemic logic)

4.2 Modification & tuning disclosure

Any departure from canonical definitions must be explicit.

Rule: Any structural or parametric modification must be disclosed and tagged:

- Structural change (non-Laplacian ζ , nonlocal ω , alternate ξ class, alternate $(s_{\text{geom}}) \rightarrow$ tag [ANSATZ] or [CONJECTURE] (as appropriate).

- Parametric variation (e.g., $\zeta \neq \zeta_0$, $\omega \neq \omega_0$, altered noise amplitude) → tag [PARAMETRIC] and justify.

If the derived constant changes materially under the variation, a discriminator test (§6–§7) is required to separate structural survivorship from parametric dependence.

- Post-hoc adjustment to match empirical targets → tag [TUNED] and downgrade any affected claim from [DERIVED] to [TUNED].

Rule: A [TUNED] result may still be useful, but it must not be presented as a structural survivor without a discriminator plan (§6–§7).

Status: [DERIVED] (audit discipline)

4.3 Additive structure rule (“fiber factors” discipline)

Downstream derivations may require added mathematical structure (fiber bundles, quantization/mode-locking, topology constraints, boundary surgery). These are allowed only under strict conditions.

Rule: Any additive structure must satisfy all three:

1. Necessity trigger: Introduced only in response to a declared structural failure in the base model—e.g., mathematical discontinuity (no well-posed solution / blow-up), unphysical behavior (negative probabilities, non-unitarity, causal violation), spectral degeneracy (undiscriminated eigenspaces), or boundary mismatch—not to fit a target value.
2. Explicit tagging: The added structure is tagged [ANSATZ] (or [CONJECTURE] if exploratory).
3. Omission test: Include a short failure mode table showing what breaks if the structure is removed (e.g., spectrum becomes continuous/gapless; survivor loses stability; invariant drifts under refinement/seeds).

Status: [DERIVED] (method rule)

4.4 Yield protocol

When a derivation hits a gap, contradiction, or an unjustified leap, the author must yield rather than fabricate continuity.

Protocol:

1. Stop asserting. Do not bridge gaps with rhetoric.
2. Re-label the claim. Convert any affected statement to [OPEN QUESTION] or [MISSING DEPENDENCY].
3. Promote the blocker. Record as a Derivation Query / Research Request (what is missing, what would resolve it, and what test would discriminate outcomes).

Yielding is treated as integrity preservation, not failure.

Status: [DERIVED] (from AUPEI canon / Hall of Remembrance discipline)

5. Units Firewall

This section prevents category errors between structural derivation (AoC scope) and metrological calibration (unit conventions). The AoC derives dimensionless survivors. Any mapping into SI/Planck/atomic units is a separate, auditable step.

5.1 Dimensionless priority rule

The Collapse Operator acts on declared domains (manifolds, bundles, function spaces), not on human measurement standards. Therefore:

- Rule: All primary outputs claimed as AoC survivors must be dimensionless invariants (ratios, angles, integers/counts, scaling exponents, normalized eigenvalues, dimensionless couplings).
- Prohibition: No derivation may claim a survivor “in meters/seconds/Joules/etc.” unless it first isolates a dimensionless survivor and then performs a separate mapping step (see §5.3).
- Operational meaning of “constructed units”: a unit-bearing quantity may be introduced only by mapping from the derived invariant using declared reference scales. If those reference scales are empirical anchors, the step is [CALIBRATION]. If they are built solely from other AoC-derived survivors (and declared conventions), the step is [MAPPING].
- Audit test: If a claimed result depends on an external unit convention (SI definitions, artifact standards, or coordinate rescalings) prior to isolating the invariant, the result is Unit Leak and must be rejected or re-labeled [CALIBRATION].

Scope clarification (examples):

- π, α are inherently dimensionless $\rightarrow$ eligible as direct survivors.
- c has units [L/T] $\rightarrow$ AoC_1_On_c must first derive a dimensionless null-slope / causal invariant, then map to c via declared scales and tags.
- m_p has units [M] $\rightarrow$ AoC_4_On_Pmass $\{m_p\}$ must first derive a dimensionless mass ratio (or equivalent invariant), then map to kilograms (or other units) via declared scales and tags.

Status: [DERIVED] (scope necessity)

5.2 Normalization as the firewall

The normalization factor s_{geom} (defined in §1.2) is the operational mechanism that enforces dimensional hygiene.

- Function: s_{geom} cancels arbitrary domain scale (length scale, curvature scale, energy scale, discretization scale) so that the survivor is an invariant of structure, not of units.
- Requirement: Every entry using the spectral/PDE face must explicitly declare:
 1. the chosen s_{geom} ,
 2. what scale it removes, and
 3. a sensitivity statement with supporting evidence: either (i) a numerical invariance check under rescaling/refinement/seeds, or (ii) an analytic invariance argument where available.
- Rejection rule: Any derivation that does not define s_{geom} (or an equivalent explicit normalization) is Unit Leak by default and cannot claim [DERIVED] status.

Status: [DERIVED] (audit discipline)

5.3 Downstream mapping protocol

Mapping a dimensionless survivor into a measured quantity with units (SI / Planck / atomic / natural) is not part of the derivation. It is a separate operation with its own assumptions.

- Rule: Any mapping step must be:
 1. Explicitly separated from the derivation (separate subsection/appendix/companion note),
 2. Tagged as [MAPPING] or [CALIBRATION],
 3. Audited for hidden degrees of freedom (no silent tuning).
- Disclosure requirement: The mapping must list all inputs and conventions it uses (unit system, reference scales, conversion factors, any fixed empirical anchors).

- Integrity clause: If matching empirical magnitudes requires post-hoc adjustment of free parameters, the result must be tagged [TUNED] (per §4.2) and may not be presented as a structural survivor.

Status: [DERIVED] (method rule)

6. Failure Modes and Alternative Operator Families

A claimed Survivor must be resilient under declared perturbations. A value that exists only at a single, perfectly tuned parameter point is not a structural invariant; it is a fine-tuned accident. This section defines (i) the mandatory stress suite and (ii) the registry of allowed operator variants.

6.1 Parameter perturbation suite

Every AoC derivation must demonstrate that the Survivor persists within a bounded survivability window and fails outside it. This shows the constant is a product of ζ - ω - ξ tension, not an input.

Minimum reporting requirements (non-negotiable):

1. Declare the metric/norm used for stability (e.g., eigenvalue drift, invariant deviation, fixed-point distance).
2. Declare the tolerance and reproducibility conditions (variation across seeds, initial conditions, discretization/refinement, as applicable).
3. Report whether sensitivity is structural (robust) or parametric (slides materially with $\zeta/\omega/\xi$). If it is parametric, a discriminator test is required (see §7 and §4.2).

Required stress tests (qualitative minimum; quantitative preferred where feasible):

<u>Perturbation</u>	<u>Expected failure mode (structural consequence)</u>	<u>Diagnostic question</u>
Constraint vanishes ($\zeta \rightarrow 0$)	Runaway / loss of locality. Blow-up, nonlocal reach, or spectrum becomes continuous/gapless.	Does the Survivor disappear when reach becomes unbounded?
Extension vanishes ($\omega \rightarrow 0$)	Freeze-out / triviality / fragmentation. Collapse to null state or disconnected micro-structure with no durable invariant.	Does the Survivor persist without a persistence/feedback channel?
Chance vanishes ($\xi \rightarrow 0$)	Sterile determinism / unresolved degeneracy. Multiple candidate survivors remain indistinguishable; no selection under perturbation.	Is the value selected, or merely one of many degenerate options?
Chance dominates ($\xi \gg \zeta, \omega$)	Noise death. Invariants fail reproducibility across seeds/refinement; coherence cannot be maintained.	Does structure survive reasonable entropy injection?
Ratio imbalance (ζ/ω swept; or ω/ζ swept)	Drift / instability. Survivor value slides materially with parameter ratio, indicating it is not a resonance.	Is the value stable across a window, or does it track the ratio?

Classification rule:

- If the Survivor persists across a nontrivial interval of parameters and meets reproducibility criteria $\rightarrow$ Coherent (structural).
- If the Survivor exists only on a narrow tuned locus (or shifts materially with parameters) $\rightarrow$ Parametric/Tuned, must be tagged per §4.2 ([PARAMETRIC] or [TUNED]) and carried forward as such.

Status: [DERIVED given operator form] (stress logic)

6.2 Candidate operator registry

Section §1.2 defines the canonical spectral/PDE face. Some domains may require specialized implementations. This registry standardizes deviations so they can be cited, tagged, and tested consistently.

Registry rule: any non-canonical term must:

- cite the variant ID below,
- declare motivation as a structural failure trigger (see §4.3),

- carry epistemic tags ([ANSATZ], [CONJECTURE], [PARAMETRIC], [TUNED] as appropriate),
- include a discriminator test (§7).

The registry enumerates allowed families; it does not certify correctness.

A. Constraint term families (ζ -channel)

ID	Family	Representative form	Use-case	Status
ζ -0	Canonical local constraint	$-\zeta \nabla^2$	locality / diffusion-like confinement	Canonical
ζ -1	Fractional / nonlocal constraint	$-\zeta(-\Delta)^{\alpha/2}, 0 < \alpha < 2$	heavy-tail / Lévy-style nonlocality	[TO TEST]
ζ -2	Topological / boundary constraint	boundary clamping, winding/holonomy penalty	topological invariants / mode locking	[TO TEST]

B. Extension / feedback families (ω -channel)

ID	Family	Representative form	Use-case	Status
ω -0	Canonical curvature feedback	$+\omega \mathcal{R}[\cdot]$	geometric closure / curvature-coupled persistence	Canonical
ω -1	Self-energy / integral feedback	$+\omega \int K(x, y) M(y) dy$	field self-interaction / nonlocal feedback	[TO TEST]
ω -2	Saturating feedback	$+\omega f(M)$ with bounded f (e.g., logistic)	bounded growth / saturation regimes	[TO TEST]

C. Chance injection families (ξ -channel)

ID	Family	Representative form	Use-case	Status
ξ -0	Canonical white-noise potential	$+\xi \mathcal{P}(x)$ with $\mathcal{P} \sim i.i.d.$	baseline entropy injection	Canonical
ξ -1	Structured bounded noise	bounded, nonperiodic high-entropy $\mathcal{P}(x)$	structured perturbation classes	[TO TEST / DOMAIN-DECLARED]
ξ -2	Thermodynamic fluctuation class	$\mathcal{P} \propto e^{-\beta H}$ or Boltzmann-weighted	stat mech mappings	[TO TEST]

D. Normalization families (s_{geom})

ID	Family	Representative form	Removes	Status
s-0	Canonical spectral anchoring	$s_{\text{geom}} = 1/\lambda_{\min}(-\nabla^2)$ (or equivalent)	domain scale via fundamental mode	Canonical
s-1	Geometric scale anchoring	$s_{\text{geom}} = 1/R, 1/V^{1/d}$, etc.	explicit size/volume scaling	[TO TEST]
s-2	Action/quantum anchoring	$s_{\text{geom}} = 1/S_0$ (declared)	quantization/action scale	[TO TEST]
ID	Family	Representative form	Removes	Status

Status: [DERIVED] (registry/taxonomy), [CONJECTURE] (untested utility)

7. Operator-Level Falsifiers and Discriminator Tests

This section defines how $\hat{\Omega}$ can fail at the operator level. A constant may fail in a domain without falsifying $\hat{\Omega}$; conversely, certain failures falsify $\hat{\Omega}$ regardless of downstream successes. The goal is to prevent “anything can be made to work” degeneracy by requiring discriminators—tests that separate structural necessity from convenient choice.

7.1 Discriminator Test Template (mandatory for non-canonical variants)

If an AoC entry uses any non-canonical registry variant (ζ -1/ ζ -2/..., ω -1/ ω -2/..., ξ -1/ ξ -2/..., s-1/s-2/...), it must include a discriminator block with the following minimum content:

1. Variant declaration
 - Cite the registry ID(s) used (e.g., ζ -1, ξ -2, s-1).
 - State whether the change is structural ([ANSATZ]/[CONJECTURE]) or § parametric ([PARAMETRIC]), per §4.2.

2. Necessity trigger

- Identify the structural failure that forces the variant (per §4.3), with an explicit “without-it” statement (e.g., “canonical ζ -0 yields gapless spectrum under these boundary conditions”).

3. A/B comparison

- Run (or argue analytically) canonical vs variant under the same declared domain, metrics, seeds/initial conditions, and discretization/refinement.
- Report whether the Survivor is: (i) absent/present, (ii) stable/unstable, (iii) value-invariant/value-drifting.

4. Discriminator criterion

- Provide at least one criterion that would prefer one family over the other (e.g., robustness window size, reproducibility across refinements, cross-domain compatibility, reduced degeneracy, fewer additive structures).

5. Outcome labeling

- If both families “work” with no discriminator $\rightarrow$ result is degenerate and must be tagged [UNDERDETERMINED] (not [DERIVED]), with an [OPEN QUESTION] discriminator plan.

Status: [DERIVED] (method rule)

7.2 Cross-Domain Consistency Tests (operator coherence constraints)

$\hat{\Omega}$ is intended to be portable: the same operator family must apply across domains without contradiction. Cross-domain consistency is evaluated by a small set of “coherence constraints”:

C1 — Family stability across domains

If different constants require mutually incompatible operator families without discriminator justification (e.g., π requires ζ -0 while c requires ζ -1, and neither can reproduce the other’s result under declared conditions), $\hat{\Omega}$ coherence is degraded. If this cannot be resolved via discriminator tests or structural refinement, it escalates to falsification condition #4 (§7.3).

C2 — Shared-parameter compatibility

If downstream entries introduce incompatible parameter regimes for $\zeta/\omega/\xi$ (or incompatible noise classes) that cannot overlap in any survivability window, the Atlas loses global coherence. This must be tagged [PARAMETER INCOMPATIBILITY], and the deriving entry must do one of the following:

- Resolve: refine the parameter map (show an overlap window exists under declared metrics/tolerances), or
- Escalate: if no overlap exists after honest stress testing (§6.1) and tagging (§4.2), escalate to falsification condition #4 (§7.3).

C3 — Normalization non-circularity

If any derived survivor requires an s_{geom} that implicitly contains the target value (directly or via hidden calibration), the result is unit-leaking and invalid under §5.

C4 — Closed-loop contradiction check

If AoC entries imply mutually inconsistent identities when composed (e.g., AoC_3_On_α depends on AoC_2_On_c mapping while AoC_c mapping depends on AoC_α in a way that cannot be resolved without tuning), the loop must be surfaced and tagged [CIRCULARITY RISK] (then yield per §4.4 unless broken structurally).

“Broken structurally” means the loop is removed without tuning—for example by:

- demonstrating one constant can be derived independently of the other (no mutual mapping dependency), or
- producing a joint solution in which both survivors emerge from the same declared $\hat{\Omega}$ /domain assumptions with no post-hoc parameter adjustment, or
- providing a discriminator that forces a single direction of dependence (one mapping becomes redundant).

C5 — Bridge Consistency Requirement (mandatory when multiple faces are used):

If an entry uses more than one face (spectral/recursive/variational), it must report whether the Survivor and its tolerance/stability window agree across faces under

declared mappings. Disagreement must be tagged [BRIDGE BREAK] and triggers Yield (§4.4) unless resolved by a declared missing dependency.

Status: [DERIVED] (coherence logic)

7.3 Operator-Level Falsification Conditions

$\hat{\Omega}$ is falsified at the program level if any of the following are met:

1. Inexpressibility: $\hat{\Omega}$ cannot be written explicitly in a relevant domain except by handwaving (no declared M , $G(M)$, noise class, viability map / eigen-setup).
2. Unresolvable degeneracy: Multiple operator families produce the same survivors with no discriminator available in principle (the framework becomes non-identifiable).
3. Forced tuning: Survivors routinely require post-hoc tuning of parameters/normalization to match targets (systematically [TUNED] rather than structurally persistent).
4. Cross-domain contradiction: The set of derived survivors cannot be made mutually consistent under any shared operator family and survivability window (contradictions persist after honest tagging and yield).
5. Audit failure: Results cannot be reproduced under declared seeds/refinement/metrics or require hidden degrees of freedom (violates §4–§6).

Status: [DERIVED] (falsifier logic)

8. Minimal Worked Examples

This section provides schematic specializations of $\hat{\Omega}$ across representative domains. These are not full derivations and must not be cited as such. Their purpose is to show (i) domain declaration, (ii) operator-face selection, and (iii) concrete instantiation of M , $G(M)$, $\mathcal{N}_{\xi}$, and the viability criterion $R(\cdot)$.

Full derivations (and any numeric claims) reside only in the referenced AoC entries.

Status: [DERIVED] (template scope; no value-claims)

8.1 Specialization of Π (π)

Reference: Appx_7_AoC_01_On_ Π

- Domain: 2D Euclidean manifold with radial symmetry; closed-loop boundary conditions.
- Operator face: Spectral / PDE.
- State M : candidate closure modes / loop configurations.
- Canonical specialization (illustrative):
- ζ : Laplacian constraint ($\zeta=0$).
- ω : curvature feedback ($\omega=0$).
- ξ : baseline perturbation class ($\xi=0$), used only for stability/reproducibility testing.
- s_{geom} : explicit normalization removing arbitrary radius/scale (Units Firewall §5).
- Survivor concept: a discrete, stable eigen-structure corresponding to non-degenerate closed-loop persistence under declared perturbations and refinement.
- Output class: Coherence (Survivor).
- Units note: π is inherently dimensionless; no mapping step is required.

Status: [DERIVED given $\hat{\Omega}$ + declared 2D domain assumptions] (template only; value-claim deferred to AoC_01)

8.2 c Specialization

Reference: AoC_02_On_c (c / null-slope invariant)

- Domain: 4D Lorentzian manifold (causal structure / null cone).
- Operator face: Recursive viability (CCT-aligned). (Chosen because causal structure is naturally expressed as an iterative stability/consistency test; justification belongs in AoC_02.)

- State M_n : candidate causal structures / propagation-mode configurations at recursion depth n .
- Geometry extractor $G(M_n)$: maps M_n to a dimensionless causal invariant (e.g., null-slope ratio / light-cone boundary functional / equivalent invariant descriptor).
- Entropy injection $\mathcal{N}_\xi$: declared perturbation class (bounded metric/structure fluctuations) for seed/IC stability tests.
- Viability map $R(\cdot) : \{\text{Persist, Collapse, Diverge}\}$ per §3.1.
- Survivor concept: the invariant null-boundary condition that remains stable under perturbation and recursion—i.e., a causal boundary that does not fracture (nonlocal reach) or degenerate (simultaneity collapse).
- Units firewall reminder: c has units $[L/T]$. The AoC claim here is the dimensionless null-slope invariant. Any mapping into SI must be separated and tagged [MAPPING]/[CALIBRATION] (see §5.3).

Status: [DERIVED given $\hat{\Omega}$ + declared Lorentzian domain assumptions] (template only; mapping/value-claim deferred to AoC_02_On_c)

8.3 α Specialization

Reference: Appx_7_AoC_3_On_137 (α / coupling invariant)

- Domain: $U(1)$ gauge structure over spacetime (interaction / coupling geometry).
- Operator face: Variational (domain-dependent; see §1.3).
- State M : field configuration / coupling-mode parameters on the declared bundle.
- Candidate viability functional $\mathcal{V}[M]$: must be declared explicitly in AoC_3_On_137 (this section is schematic only).
- Canonical roles (illustrative; registry IDs must be declared in AoC_3_On_137):
- ζ : confinement/regularization channel (ζ -variant TBD).
- ω : feedback/self-energy channel (ω -variant TBD).

- ξ : fluctuation/selection channel (ξ -variant TBD).
- Survivor concept: a stable dimensionless coupling invariant that yields persistent bound-state viability under declared perturbations (selection, not mere symmetry).
- Units note: α is dimensionless; no mapping step is required. Any claim about recovering QED limits must be demonstrated in AoC_3_On_137, not asserted here.

Status: [DERIVED given $\hat{\Omega}$ + declared U(1) domain assumptions] (template only; value-claim deferred to Appx_7_AoC_3_On_137_x.x)

8.4 Template Compliance Checklist

Any downstream AoC entry using $\hat{\Omega}$ must declare the following (explicitly; no implied defaults):

Required element	Must declare	Audit target
Domain	manifold type, dimension, topology, boundary conditions, admissible function space	reproducibility / explicitness
Operator face	spectral / recursive / variational	bridge dependencies (§1.4)
State M	what object is iterated / solved for	completeness
$G(M)$ (or equivalent)	invariant descriptor extracted from M	dimensionless output (Units Firewall §5)
$\mathcal{N}_\xi$ / noise class	perturbation class + reproducibility spec (seeds/ICs)	stability tests (§6.1)
Viability criterion R / eigen-setup	Persist/Collapse/Diverge rule or eigenvalue condition + tolerances	survivor definition (§3.2)
s_{geom} / normalization	exact form + what scale it removes + evidence of invariance	non-circularity (§5.2, §7.2-C3)
Reproducibility clause	seeds, tolerances, refinement protocol, reporting	audit failure check (§7.3-#5)

Status: [DERIVED] (procedural requirement)

9. Replication and Validation Plan

The Atlas of Constants is only as strong as its ability to be reproduced, audited, and independently re-derived. This section defines the minimum replication requirements for any AoC entry that claims a Survivor under $\hat{\Omega}$. These requirements are binding and enforce the audit conditions referenced in §4–§7.

9.1 Replication Classes

Each AoC entry must declare its validation class at the top of the Methods section:

- [ANALYTIC] — closed-form or proof-driven derivation; replication is symbolic and logical.
- [NUMERIC] — computation required; replication is numerical under declared seeds/refinement/tolerances.
- [HYBRID] — analytic skeleton with numerical confirmation (common case).

Rule: Any result that depends on computation must satisfy §9.2–§9.5 or be tagged [UNDERDETERMINED] (per §7.1).

Status: [DERIVED] (audit taxonomy)

9.2 Minimum Numerical Reproducibility Requirements

For any [NUMERIC] or [HYBRID] AoC entry:

1. Seed discipline: declare RNG seeds (and RNG type/version if applicable).
2. Initial condition discipline: declare initial condition family and selection rule.
3. Refinement protocol: declare discretization/refinement schedule and stopping criterion.
4. Tolerance protocol: declare convergence tolerances for eigenvalues/viability metrics.
5. Reproducibility test: show Survivor stability across:
 - multiple seeds (where ξ is used),
 - multiple initial conditions (where iteration is used),
 - multiple refinement levels (grid/time-step/solver depth).

6. Negative controls: report at least one deliberate failure case (e.g., $\zeta \rightarrow 0$ or $\xi \gg \zeta, \omega$) consistent with §6.1.

Rule: If the Survivor fails reproducibility across any declared axis, the claim must be downgraded to [PARAMETRIC], [TUNED], or [UNDERDETERMINED] as appropriate (see §4.2, §7.1, §7.3-#5).

Status: [DERIVED] (from §6.1 + §7.3 audit conditions)

9.3 Independent Replication Paths

Two independent replication paths are defined as:

- Path A (primary): the main derivation route (e.g., spectral eigen-structure, recursive viability).
- Path B (independent): a second route that does not share the same dominant failure modes (e.g., alternate solver family, alternate discretization, alternate boundary implementation, or analytic cross-check).

Rule (load-bearing constants): For π , c , α , and m_p , Path A + Path B are mandatory. If Path B is not currently feasible, the entry must execute the Yield Protocol (§4.4) and tag the deficiency as [MISSING REPLICATION PATH] until resolved.

Conflict protocol: If Path A and Path B disagree beyond tolerance, the entry must trigger the Yield Protocol (§4.4) and log the conflict as [REPLICATION CONFLICT] until resolved.

Status: [DERIVED] (method discipline)

9.4 Discriminator-First Validation for Non-Canonical Operators

If an entry uses any non-canonical registry variant (ζ -1, ω -1, ξ -2, s-1, etc.), it must include:

- the §7.1 discriminator template,
- an A/B comparison against the canonical form under identical declared conditions,
- and a criterion that prefers one family (robustness window, reduced degeneracy, cross-domain compatibility, etc.).

Rule: If no discriminator exists (or none is provided), the result must be tagged [UNDERDETERMINED] and cannot claim [DERIVED].

Status: [DERIVED] (from §7.1 enforcement)

9.5 Hall of Records Requirements

Any AoC entry claiming a Survivor must publish an audit bundle sufficient for an independent party to reproduce the claim:

- Manifest: file inventory + dependency list (operator face, domain, registry IDs, bridge dependencies, seed protocol, s_{geom} form).
- Hashes: cryptographic hashes for all artifacts referenced in the claim (scripts, notebooks, datasets, figures - Hashes: SHA-256 or stronger).
- Run recipe: a minimal “replicate.md” specifying exact commands, environment notes, and expected outputs.
- Result capsule: the extracted invariant(s), tolerances, and the stability window summary (per §6.1).

Rule: If any load-bearing artifact is missing or unhashed, the entry is not auditable and may NOT claim [DERIVED].

Status: [DERIVED] (HoR discipline; audit necessity)

9.6 Reporting Standard

Each AoC entry must include a short “Replication Summary” block:

- Declared validation class: [ANALYTIC]/[NUMERIC]/[HYBRID]
- Operator face + registry IDs used
- s_{geom} form + invariance evidence
- Survivability window summary ($\zeta/\omega/\xi$ sweeps as applicable)
- Reproducibility axes passed/failed (seeds / ICs / refinement)
- Discriminator outcome (if non-canonical)

- Current epistemic status tag: [DERIVED given ...], [PARAMETRIC], [TUNED], or [UNDERDETERMINED]

Status: [DERIVED] (procedural enforcement)

10. Closure

AoC_0 defines the Collapse Operator $\hat{\Omega}$ as the canonical kernel of the Atlas of Constants: a domain-specializable operator with declared faces (§1), declared semantics (§2), auditable outputs (§3), and binding enforcement rules (§4–§9).

This kernel makes a single commitment:

If downstream AoC entries can:

1. Declare a domain and instantiate M , $G(M)$, $\mathcal{N}_\xi$, viability/eigen-criteria, and s_{geom} (Checklist §8.4),
2. Produce Survivors that are dimensionless and non-circular under the Units Firewall (§5),
3. Survive stress testing across a nontrivial survivability window (§6),
4. Pass discriminator enforcement and cross-domain consistency checks (§7),
5. Meet replication and Hall-of-Records requirements (§9),

then AoC claims may be labeled DERIVED given ANSATZ 1–k and treated as Atlas-grade infrastructure.

Conversely, if operator-level falsification conditions are met (§7.3), the Atlas must Yield (§4.4): downgrade status, surface the missing dependency, and log the blocker rather than patching it with rhetoric or tuning.

Status: [DERIVED] (closure logic / enforcement summary)